

Music Atlas: six people, two journeys

What a music teacher, a self-taught 15-year-old and a gigging musician found when they worked through the Ionian lessons into the Studio, and through the Studio demos back into a lesson — and what each would need before paying \$20 a month.

20 September 2026 · Webapp-Refactor, branch `Peter`

Six AI research agents, each role-playing one persona, driving a real browser against a local dev build.

Findings verified against source code wherever possible; environment artefacts are marked separately.

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1. Summary — what matters most

Every persona found something in Music Atlas they genuinely wanted. The gigging musician called the degree-number chord format *“the first time a piece of software has written chords the way I say them out loud”* (a line he later retracted after checking Hookpad, which already labels chords degree-first — but the reaction is the point: it spoke his language). The teacher said the notation engine was good enough to project and that selecting and editing a note *“behaves like notation software.”* The teenager said the lesson piano roll *“looks exactly like the beat app on my phone.”*

What stopped them most often was not a missing feature. It was that **the app rarely tells you how you did, and never tells you why you are where you are.**

The single most important finding. The play-along activity marks itself complete when the playback reaches the end — regardless of what was played. Two personas independently tested this: one played eight deliberately wrong notes, one played nothing at all. Both were told *“Nice work! You finished the play-along”* and given a completion tick. The app is already recording every note played and drawing wrong notes on screen; the completion check simply never looks at them. Confirmed in code: `PlayAlong.tsx:273`.

The five things to take away

1. **Nothing is graded, and nothing is acknowledged.** Wrong notes produce no response; an empty performance passes. A teacher cannot assign work that marks itself complete, and a student who is lost gets silence.
2. **Nothing closes the loop.** Each activity ends with a tick and a percentage. No persona was ever told what they had just learned, or connected back to the song they came from. The musician said he learned the word *Dorian* because he stopped to read an overview page — *“not because the round trip taught it to me.”*
3. **Both bridges between Learn and the Studio exist, and both are nearly invisible.** Studio-to-Learn is a 9-pixel chip labelled with the mode name — the very word the user doesn't know. Learn-to-Studio is a card worded like a worksheet (*“Apply the chords you just learned”*), which opens a professional DAW with no first step.
4. **The Insight panel is the product's best asset, and part of what it says is wrong.** Every major chord is described as *“built on the 1st degree”* — including chords labelled 4 and 5 two lines above. Progression matches all score 100% without matching the song's actual progression.
5. **Work doesn't persist.** There is no link that reopens a project; one persona renamed his project, followed a lesson link, and came back to find the rename gone. A teacher cannot link a class to a specific step.

How each persona answered the question they were sent to answer

PERSONA · JOURNEY DIRECT ANSWER

Teacher Learn → Studio	Practice Track prompt was <i>“mildly enticing”</i> — exactly the Friday backing track she wants — but the hand-off <i>“lost me the instant the page loaded. I was promised a track and given a studio.”</i>
Teenager Learn → Studio	Clicked it because it was the only glowing button. The copy read as homework. In the Studio: <i>“no first step”</i> — would have quit about 60 seconds in.
Musician Learn → Studio	Couldn't see the prompt live (test environment). The Studio closed the gap anyway: the degree-number chord ruler reading <i>“1 maj / 4 maj / 5 maj”</i> over a B $\flat$ progression was <i>“literally how I talk.”</i>
Teacher Studio → Learn	<i>“Good bait, wrong hook.”</i> Insight enticed her into a lesson, but Insight showed her harmony and the lesson taught a scale — <i>“a different app.”</i>
Teenager Studio → Learn	Nothing in Insight pulled him in; the only parts in his language (VIBES, STYLES) are at the very bottom. The lesson felt like class, <i>“but a class I didn't hate.”</i>
Musician Studio → Learn	<i>“Yes on naming, no on the bridge.”</i> Insight named the music in his language; the bridge broke at the door (a bare mode name) and at the exit (silence after a correct performance).

2. Method, and what this research could not see

The personas

PERSONA	BRIEF
Music teacher	Teaches general music and beginning piano to 7th–9th graders; 14 students, 45-minute periods, a cart of keyboards (and, it emerged, Chromebooks). Reads music fluently, has engraved in MuseScore/Finale. Evaluating the app to build a unit around it. Skeptical of edtech that gamifies without teaching.
Self-taught teenager	15. A year of YouTube tutorials on a cheap MIDI keyboard. Makes beats in a phone app. Knows "C major", has never heard the word "mode", cannot read notation. Skips instructions; embarrassed by jargon.
Gigging musician	34. Fifteen years of bass and keys in wedding, funk and soul bands. Excellent ears and hands; thin theory vocabulary, and quietly self-conscious about it. Wants the <i>names</i> for what his hands already do.

The journeys

Journey A — Learn to Studio. Find the Ionian material, play through the activity flow with a (simulated) MIDI keyboard, decide whether to take the Practice Track prompt, follow it into the Studio.

Journey B — Studio to Learn. Open a demo project, use Create, Master, Score and Lead Sheet, look at Insight, decide whether to follow it into a Learn activity, do the activity.

Each persona walked both journeys, as separate agents, for six runs in all. Each agent was told to stay in character while using the app and to report as an observer of itself: quote the on-screen wording that caused a problem, classify every finding (confusion, frustration, missed, bug, question), and grade its severity.

How their process was reconstructed

The agents' full transcripts are too large to read directly. Their process was reconstructed from what they left behind: **124 numbered scripts** (each a step: where they went, what they clicked, what they waited for, what they retried) and **209 screenshots**. Repeated attempts at the same target are visible in the scripts and are reported below as friction.

What this research could not see

Environment limits — read these before weighing any finding.

- **No real account.** The app requires sign-in, so the research ran against a second dev server with the built-in development auth bypass enabled. Backend calls therefore returned 401 : XP stayed at 0, streaks and progress did not save, and the user was named "Dev Bypass".
- **The Chords chapter never loaded.** Its data comes from a backend endpoint that 401'd, so every Learn-first persona went straight from Melody to the end-of-lesson card. **Nobody saw the Melody→Chords Practice Track interstitial.** The end-of-lesson Practice Track card was seen by two of the three.
- **No sound.** Headless browsers produce no audio. Every sound-related judgement is about what the screen showed.
- **The Studio never heard the simulated keyboard.** Learn reads MIDI directly; the Studio uses the webmidi.js library, which didn't recognise the simulated device. So "NOW PLAYING — Play a chord to see analysis" never responded for any persona. **This is a test artefact, not a product bug,** and needs re-testing with a real keyboard.

3. The six runs

Teacher • Learn → Studio

23 steps 38 screenshots 92 browser actions — the most thorough run

Path. Rejected the home screen's *"Start your first lesson → Browse"* because it points at genre pathways, and went to the Theory tab instead. Found *"Ionian (Major)"* easily. Read all four Overview sub-tabs (Scale, Triads, 7th Chords, Inversions) before touching an activity. Played all six Melody steps — and ran two deliberate controls: a run of wrong notes, and one play-along where she touched nothing at all. Went straight from Melody to *"Great work! You completed the ionian lesson in C"*; took the Practice Track at Level 1; spent real time in the Studio's Score and Lead Sheet; checked the `/teacher` area.

What stopped her. A play-along completed with nothing played. Wrong notes producing no response of any kind. *"Ionian Intervals: 1, 2, 3, 4, 5, 6, 7"* — those are scale *degrees*, a word her students would have to unlearn. The next lesson offered is D $\flat$ (five flats). All six Melody steps share one URL, so she can't assign step 4 or recover from a refresh. In the Studio, the only instruction mentions *"⌘-drag"* — her students' Chromebooks have no ⌘ key.

"A student can complete a Play-Along by doing nothing. I cannot assign or grade work that marks itself complete."

Teacher • Studio → Learn

21 steps 42 screenshots — the most of any run

Path. Chose First Light because *"1-5-6-4 is a unit I already teach."* Oriented in Create, pressed Play, then gave the Score a proper engraver's reading: turned on chord symbols, selected a note, raised it with ↑, added a staccato and a sharp, undid. Hunted the File menu for Print; found none. Went to Lead Sheet, cycled the chord-format switcher, turned Melody on, exported MusicXML (it downloaded) and tried PDF. Judged Master. Woke Insight by clicking a chord, followed the *"C Ionian"* chip into a lesson and did an activity.

What stopped her. No Print or PDF for the Score anywhere in the File menu; the MusicXML that exports is four bars of whole rests plus chord symbols, with no parts. A phantom fifth bar with empty staves. A staccato applied to one note landing on another. Piano voicings stacked on ledger lines below the treble staff. The Insight sentence contradicting its own label. The lesson being a scale drill when Insight had shown her harmony.

"Good bait, wrong hook."

Teenager • Learn → Studio

23 steps 32 screenshots 8 consecutive attempts to open one card

Path. Ignored the dashboard (*"charts are boring"*). Browsed Genre, then World Harmony (*"coolest-looking page in the app"*), then Theory. Clicked **Lydian first, because it was the brightest card**, then Ionian only because *"Major"* was the one word he recognised. His scripts show eight consecutive attempts (steps 3–10) to click that card: the mode cards are plain `div`s with no button or link role, so there was nothing to address. Seven more steps (11–17) went on the Melody tab. Completed all six activities, took the Practice Track, and in the Studio found PRISM and pressed Create.

What stopped him. PRISM's Create wrote chords onto his *Melody* track and rewrote the song's chords, leaving the Chords track empty, with no undo offered (Figure 6). The Hold exercises give no feedback while you play. Nothing on the Studio screen says what a Practice Track is for. *"Continue curriculum"* is *"the single most school-sounding word possible."*

"The moment I'd have quit: roughly 60 seconds in, right after double-clicking the empty Chords lane did nothing."

Teenager • Studio → Learn

22 steps 30 screenshots

Path. Picked Midnight Groove for the name. Found Play, toured Master, Score and Lead Sheet. Opened the GROOVES browser, clicked *"Trap 1"* (nothing happened), found the unlabelled icon that actually adds it, handled a tempo-change dialog, and watched the drums change. Renamed the project "DROP v1". Scrolled Insight to the bottom, then clicked the *"D Dorian"* chip into a lesson and completed an activity. Came back.

What stopped him. His rename was gone when he returned. The Learn entry point is a 60×17-pixel chip below the fold. Everything in Insight above VIBES and STYLES is *"a wall of stuff I don't know."* The Score was closed in about two seconds. Master says MASTERING and offers an empty rack. All nine sidebar icons are unlabelled — *"I never knew a Learn section existed."*

"The doing was connected to my ears; the framing was not connected to my song at all."

Musician · Learn → Studio

20 steps 37 screenshots

Path. Went to Theory, chose B $\flat$ Ionian (*"flat keys are my world"*). Played the melody activities cleanly — then ran a control: eight deliberately wrong chromatic notes in a play-along. It passed. Tried to reach the Chords chapter by deep link; it fell back to Overview. Found the Practice Track only by expanding the mode card back on the Theory tab, which reveals four chapters (*"Overview · Melody · Chords · Practice Track"*). In the Studio, toggled the chord ruler to degree numbers.

What stopped him. The play-along not listening. The Melody tab showing Overview content until you click a step dot. The six steps being findable only in the page's accessibility labels. *"Ionian"* never attached to *"major"* inside the lesson.

"Four screens of Ionian taught me less than ten seconds of that panel."

Musician · Studio → Learn

15 steps — the most efficient run 30 screenshots

Path. Opened Midnight Groove and *"knew the tune in four seconds"* from the chord lane. Played the Lead Sheet (its playhead tracks correctly) and cycled all three chord formats. Checked Master and Score. Read Insight closely, clicked the *"D Dorian"* chip on the 2 chord, and played the Dorian scale correctly in the lesson. Returned with his project intact.

What stopped him. The bridge itself: a chip labelled only with the word he didn't know, and a lesson that ended in silence after he played it right. *"Gdom13"* in the Spelled chord format. Progression matches that *"don't know my progression."* No repeat sign or section letter on a four-bar loop.

"I left knowing a word. I know it because I read it on the Overview page — not because the round trip taught it to me."

4. Findings verified in the code

Where a persona's finding pointed at something checkable, it was traced to source. These are not impressions; they are confirmed defects with a known cause.

FINDING	CAUSE	SEVERITY
Play-along passes an empty or wrong performance. Teacher A, Musician A — independently	<code>PlayAlong.tsx:273</code> — completion is <code>currentTick >= maxEventEndTick</code> : the playhead reaching the end. The component already tracks <code>notePerformance</code> and <code>wrongNotes</code> and draws wrong notes on screen; the completion check reads neither.	BLOCKER
"Built on the 1st degree" on a chord labelled 4 or 5. Teacher B, Teacher A, Musician A	<code>chordTheoryMap.ts</code> keys its descriptions by <i>chord quality</i> . Every major triad gets "1st degree" — true of its own scale — while the label two lines above states its degree <i>in the song's key</i> . Both are true in their own frame; together they contradict each other. The sentence has to be computed from the key, not looked up by quality.	MAJOR
"Gdom13" printed as a chord symbol. Musician B	<code>leadSheetUtils.ts</code> <code>JAZZ_MAP</code> has entries for <code>dom7</code> , <code>dom9</code> , altered sevenths — but none for <code>dom11</code> , <code>dom13</code> , <code>min11</code> , <code>min13</code> or <code>maj13</code> . Unknown qualities fall through to root + internal word, so every 11th and 13th chord is affected, not just this one.	MAJOR
The Insight→Learn link is barely visible. Teenager B, Musician B, Teacher B	<code>ChordCard.tsx:147</code> — 9-pixel text, an 8-pixel book icon, and the mode name as its only label. No verb anywhere in the Studio says "learn" or "lesson".	MAJOR
A phantom fifth bar with blank staves. Teacher B · reproduced independently	Two defects in the notation engine. (1) The demos' last kick and hi-hat sit five ticks before the loop point — a humanised push into the next pass's downbeat. Quantizing snapped them onto the final barline, and <code>buildScore.ts</code> took the score's length from the snapped positions, so the piece grew a fifth bar. (2) The renderer then drew the shorter parts' fifth bar as a blank staff with no rest. Both fixed on 20 September: length now comes from what was played, and a part that has stopped is written out with a bar of rest. An earlier version of this report attributed (1) to drum hits being measured before they were clamped to their bar; the effect was described correctly, the cause was not.	FIXED

"NOW PLAYING" never responds.

Teenager A, Teenager B,
Musician A

Not a product bug. Learn reads Web MIDI directly (`useMidiInput.ts`); the Studio goes through `webmidi.js` (`MidiDeviceManager.ts`), which didn't recognise the simulated keyboard. Needs a real keyboard to judge. It does reveal the app runs **two separate MIDI input stacks**.

TEST ARTEFACT

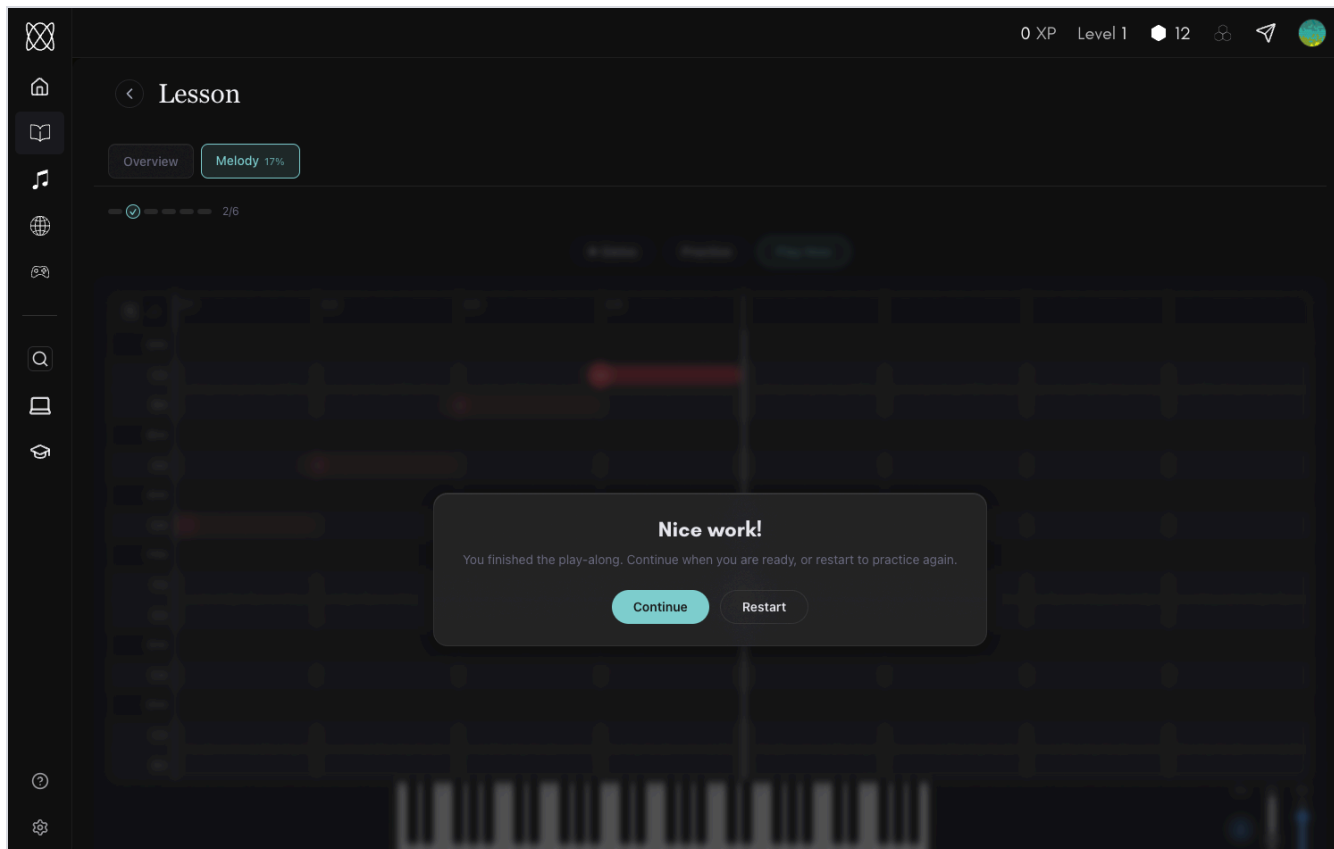


Figure 1 — A play-along completed with nothing played. Teacher A started step 2 and sat through both bars without touching a key. The modal reads “Nice work! You finished the play-along.” The step is ticked and the tab reads “Melody 17%”. The target notes are still visible, untouched, in the roll behind.

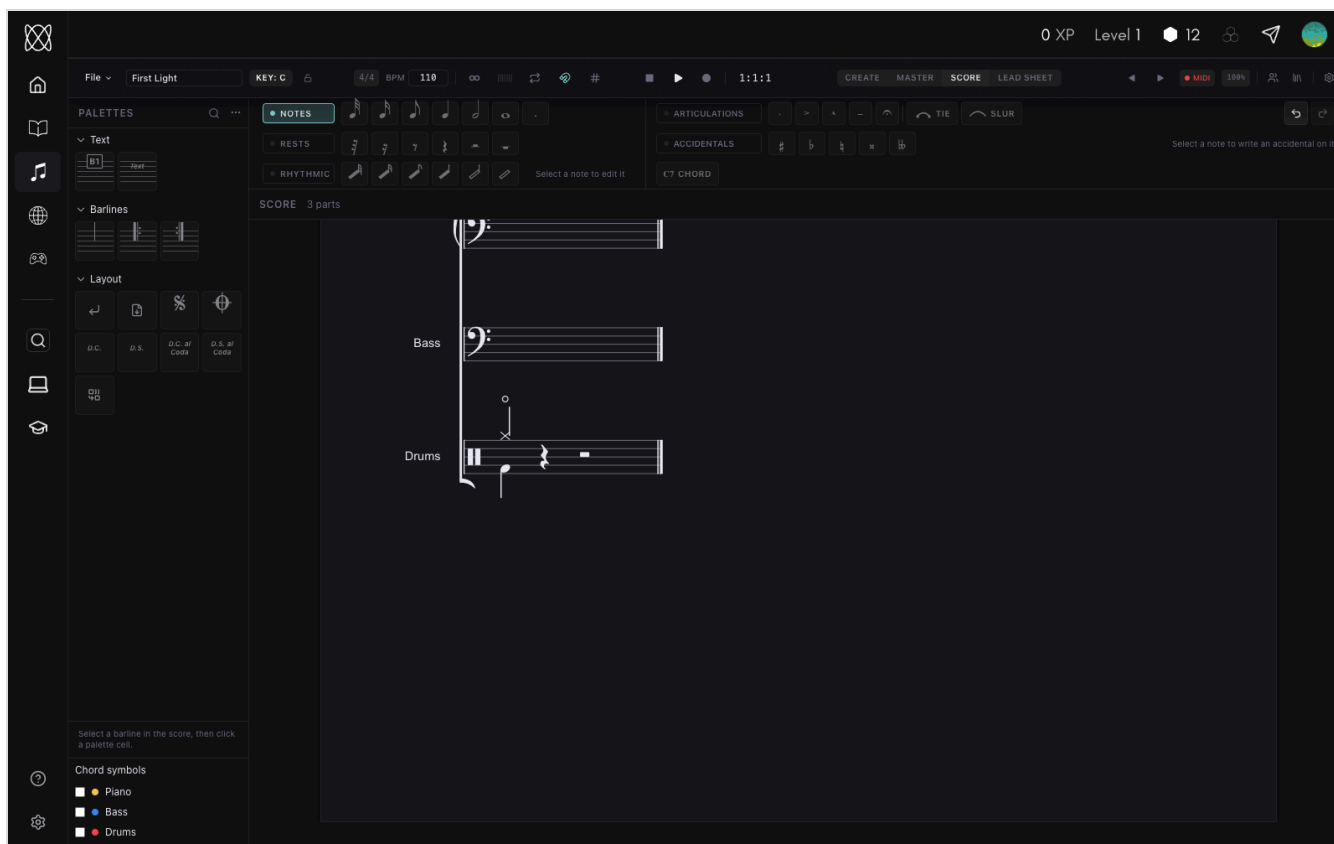


Figure 2 — The phantom fifth bar, First Light demo. The song is four bars long; the Score draws a fifth. Piano (both staves) and Bass are blank — no rests at all. The Drums bar holds a stray open-hat notehead floating above the staff with its stem crossing it, rests, and a kick hanging below. Reproduced during verification, and **fixed since**: all three demos now render exactly four bars.

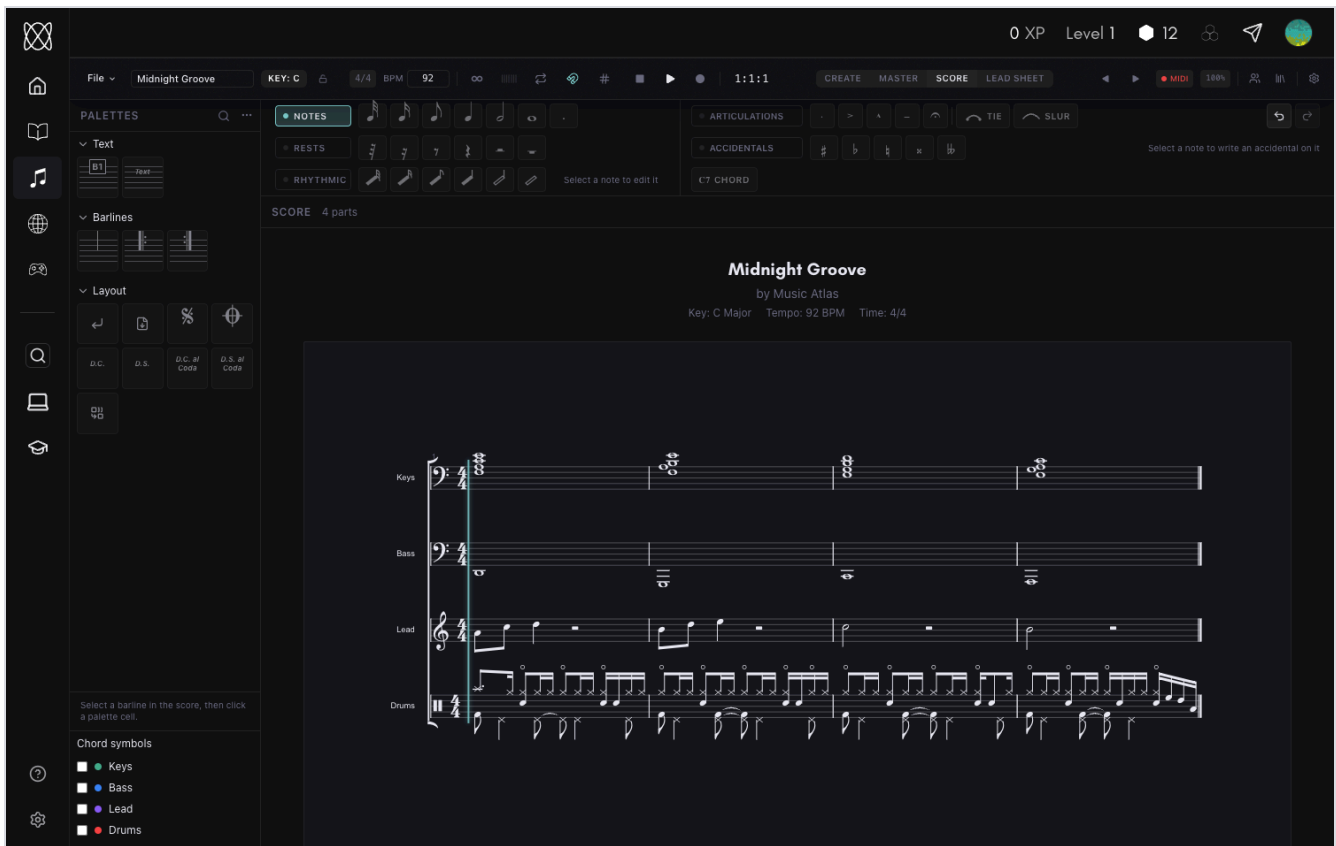


Figure 3 — The Score as a non-reader meets it (Midnight Groove). Keys are written in bass clef with the chords stacked high above the staff on ledger lines; Bass hangs below its staff; Drums runs in full notation. Chord symbols are off by default for every part. The teenager closed this view in about two seconds; the musician said “Score blocks me.”

5. Themes across personas

These are findings that two or more personas hit independently. Independent convergence is the strongest signal this research can give.

5.1 Nothing is graded, and wrong notes are met with silence 3 OF 6 · THE MOST SEVERE

Two personas ran deliberate controls and both passed an activity they should have failed. The Hold exercises give no response to a wrong note — not even lighting the key that was pressed, though the on-screen keyboard *does* light for correct notes. One persona got stuck at 15 of 16 notes on Ascend + Descend with the screen “*silently frozen*”: the exercise wants the top note struck twice, which nobody does when turning a scale around. The teacher's framing is the one that matters commercially: “*that's the whole 'gets unstuck without me' question.*”

Counter-evidence, for balance: the Studio-first teacher, who played only correct notes, praised the activity's “*immediate per-note feedback*”. The feedback that exists is for success; what is missing is feedback for failure — and grading.

5.2 Nothing closes the loop 5 OF 6

Every activity ends with a tick, a percentage, and “*Continue / Restart*”. No persona was ever told what they had just done, or why it mattered. Coming from Insight, nothing in the lesson named the song they came from; coming back, nothing in the Studio marked what they'd learned. The musician named the moment exactly: “*I played the eight notes correctly and the app said nothing. It ticked a dash, printed '17%', and slid to the next exercise. Nobody said 'that was D Dorian, that's the sound of the two chord in Midnight Groove.'*”

5.3 Both bridges are nearly invisible 5 OF 6

Studio to Learn is a 9-pixel chip reading “*D Dorian*” or “*C Ionian*” — the unknown word used as its own signpost, with no verb, sitting beside “*Parent: G Ionian*” with nothing saying which to press. Meanwhile the Library tab next door, unrelated to the user's track, has four large teal “*Start*” buttons. As the musician put it, “*the affordance is on the wrong panel.*”

Learn to Studio is a card worded like a worksheet. The teenager: “*‘Apply’ is what a worksheet says. ‘Generated’ sounds like the computer already did the fun part.*” And because the Chords chapter didn't load in this environment, the copy's “*the chords you just learned*” was false for every Learn-first persona. (That falsehood is an environment effect; the worksheet tone is not.)

5.4 The lesson and the insight teach different things 3 OF 6

Insight shows harmony — chord degrees, chord tones, progression matches. The lesson it opens teaches a scale, linearly: eight notes up a keyboard, with no chords in it. The teacher: “*the thing Insight showed me was harmonic; the lesson it sent me to is linear.*” On the G major card she was offered “*G Ionian*” and “*C Lydian*” — the latter, in a plain I–V–vi–IV pop song, would actively mislead a 13-year-old — while “*C Ionian*”, the actual key, wasn't offered at all.

5.5 "Ionian" is never tied to "major" 3 OF 3 LEARN-FIRST

The Theory card says *"Ionian (Major)"*. The moment the lesson opens, *"(Major)"* disappears: the title is "C Ionian", the body *"The key of 'C Ionian' has no sharps or flats."* The teenager's first reaction was *"is this different from C major?"* The musician: *"One sentence — 'Ionian is the major scale; you play it every night' — would have done the whole job on page one."* Ironically, World Harmony's cards all carry a one-line description (*"Bhairav — solemn dawn raga; flat 2nd & flat 6th"*); the mode cards carry none.

5.6 The Studio has no first step 4 OF 6

The teacher: *"I was promised a track and given a studio"* — four mixer strips, record-arm buttons, an FX rack and an analysis panel. The only instruction on screen reads *"click chords in the chord lane (Shift-click a run, ⌘-click to add one), or ⌘-drag across clips"*. The track the user is meant to fill is an empty black stripe *"indistinguishable from the empty space below it."* There is no way back to the lesson.

5.7 Nothing persists, and nothing has an address 4 OF 6

The scripts show all three Studio-first personas reloading the demos page 10 to 16 times, because a project lives only in memory at `/studio/editor` with no URL that reopens it. The teenager met the user-facing version: his project rename vanished after a lesson detour, with no unsaved-changes warning. The teacher met the classroom version: all six Melody steps share one URL, so she cannot send a class to step 4, and a refresh returns everyone to step 1. (Saving may also be affected by the 401s; the missing address is not.)

5.8 Insight is the best thing in the product, and it overstates its confidence 5 OF 6

Every persona praised the chord card's three registers of the same fact — interval chips, a note list, and one plain-English sentence. But the matching around it is confident and often wrong. Scale matches offer *"C Melodic Minor 89%"* and *"C Harmonic Minor 88%"* for a plain C-major song; progression matches rank ten entries, most at 100%, none of them the song's own progression (Figure 4).

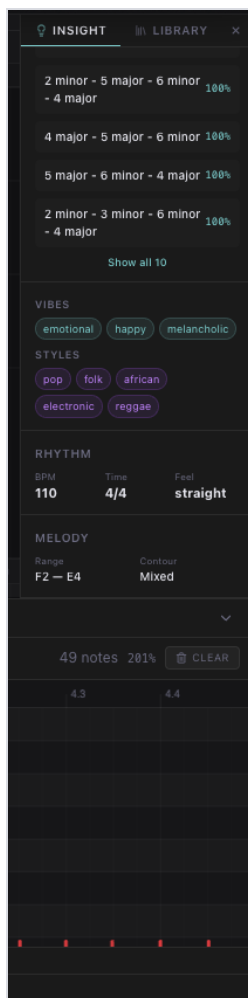


Figure 4 — Progression matches, First Light (I–V–vi–IV).

The top matches read “2 minor – 5 major – 6 minor – 4 major”, “4 major – 5 major – 6 minor”, “5 major – 6 minor – 4 major”, all at 100%. None is the song's 1–5–6–4. VIBES and STYLES, the teenagers' vocabulary, sit below them.

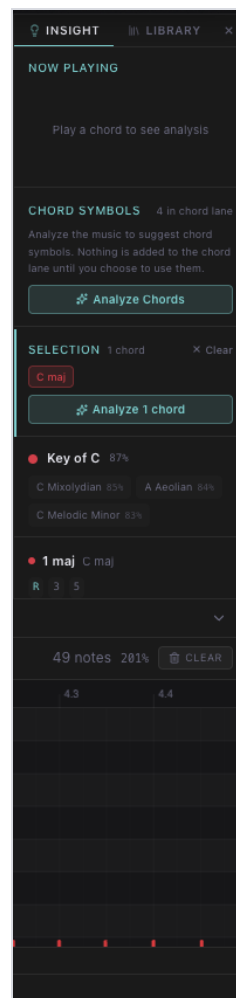


Figure 5 — The top of Insight. Three controls compete to "analyze": “Play a chord to see analysis”, “Analyze Chords”, “Analyze 1 chord” — while the analysis is already on screen. Scale matches offer “C Melodic Minor 83%” for C major.

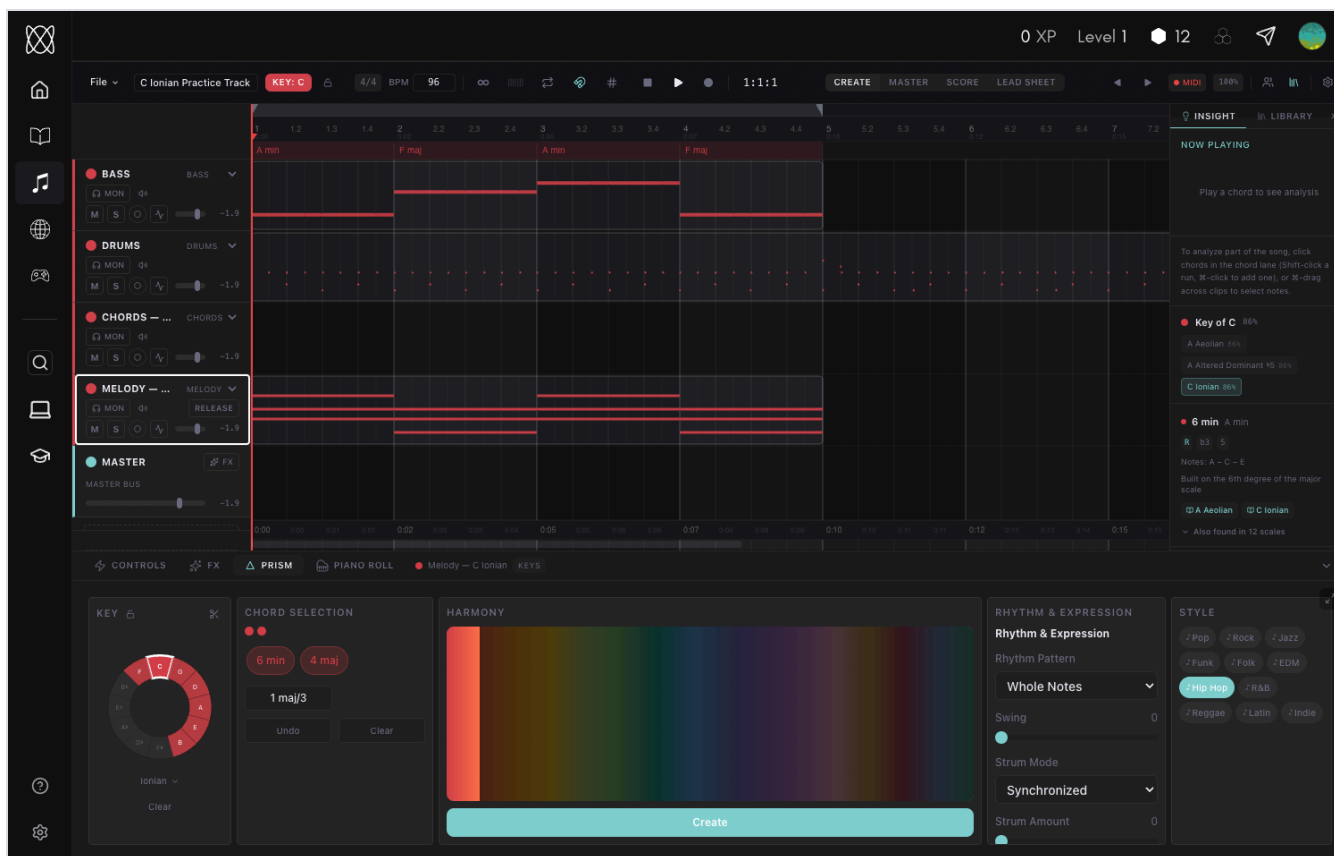


Figure 6 — PRISM's Create, after the teenager pressed it (C Ionian Practice Track). The chord lane now reads “A min | F maj | A min | F maj” — it read C / F / G / C before. The selected **Melody** track (outlined) now holds stacked three-note chords; the **Chords** track above it is still empty. The Practice Track card had promised a generated melody. No undo was offered. Reported by one persona; not yet traced in code.

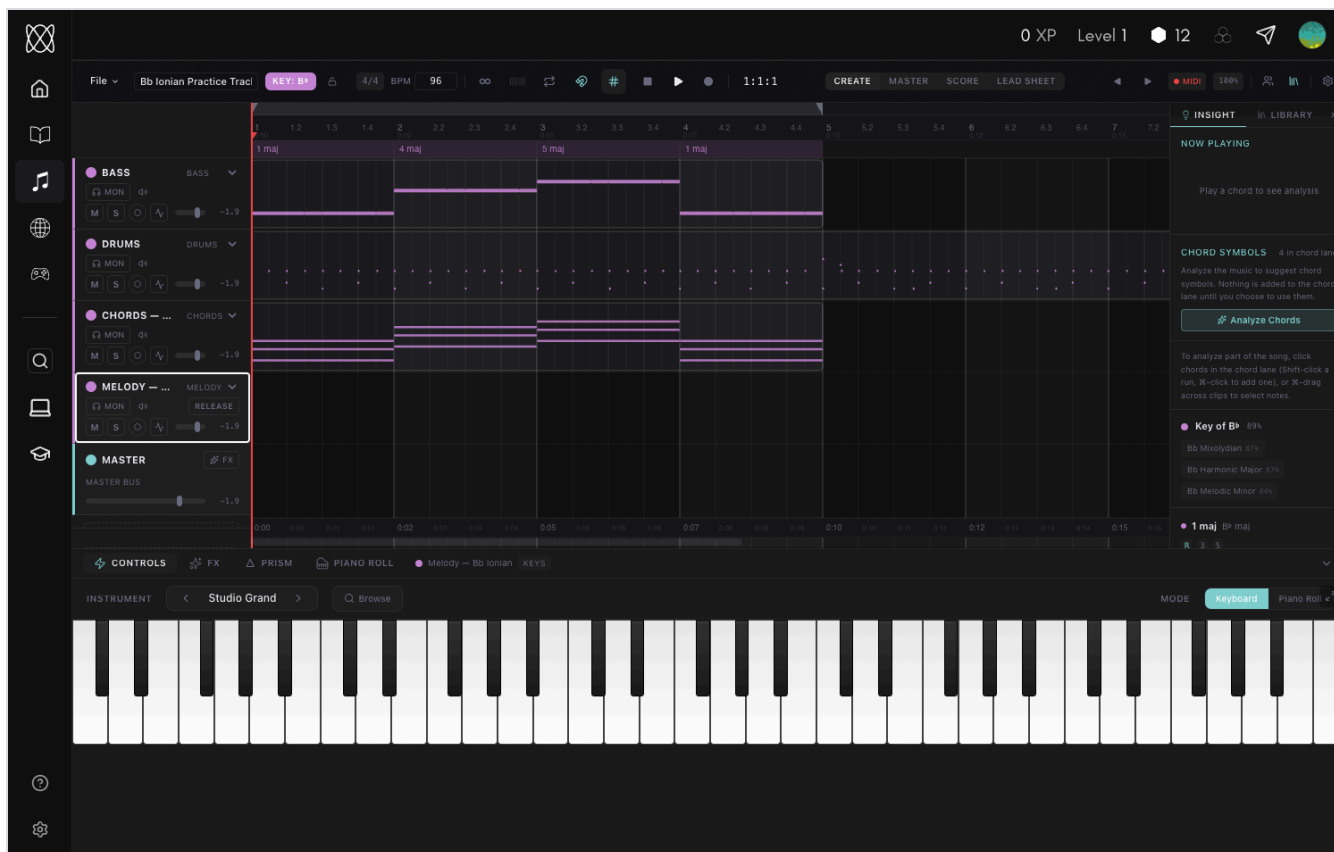


Figure 7 — What worked: the degree-number chord ruler (B $\flat$ Ionian Practice Track). With the # toggle on, the chord lane reads “1 maj | 4 maj | 5 maj | 1 maj” over a B $\flat$ progression, in the right key and tempo, with bass, drums and chords generated and the Melody track left empty for the player. The musician's single best moment in the product — found by accident behind an unlabelled button.

6. Features that are under-used or mis-used

Things that already exist and would have helped, but that personas walked past, found by accident, or used expecting something else.

FEATURE	WHAT HAPPENED	WHY IT WAS MISSED
PRISM	The teenager's favourite thing in the app — colour-wheel key picker, tappable chord chips, genre chips. <i>"The part a 15-year-old would actually play with for an hour."</i>	A collapsed bottom tab named <i>"PRISM"</i> , a word that means nothing to him, beside CONTROLS / FX / PIANO ROLL. Found by poking.
GROOVES	The other teenager's favourite — genre-tagged patterns (<i>"Trap / 808"</i> , <i>"Hip Hop / Boom Bap"</i>) with previews.	Clicking a groove's name does nothing; only an unlabelled icon adds it.
Degree-number chord ruler	The musician's best moment: <i>"1 maj / 4 maj / 5 maj"</i> over his own changes.	Behind a # button with no label. Found by accident.
Chord-format switcher (Hybrid / Numbers / Spelled)	Gives the teacher the <i>"C / G / Am / F"</i> she wanted, and the musician his degree numbers.	The button shows its <i>current</i> state with no tooltip, not what it will change to. The teacher found Spelled <i>"by clicking blindly."</i>
The six Melody steps	A perfectly good sequence: Ascend / Descend / Both × Hold / Play Along.	Rendered as six unlabelled 14-pixel dashes. Their names exist only in accessibility labels. Three personas couldn't find them; one found them by dumping the page's DOM.
Lesson chapters (Overview · Melody · Chords · Practice Track)	The clearest map of a lesson in the product.	Shown only when you expand a mode card on the Theory tab. The lesson page itself shows two tabs. The musician found the Practice Track <i>"by clicking a card chevron by luck."</i>
VIBES / STYLES	<i>"cool, sexy, hypnotic"</i> ; <i>"hip hop, neo-soul, r&b"</i> — <i>"those are my words."</i>	At the very bottom of the Insight panel, below everything the teenager didn't understand.
Lead Sheet PDF / MusicXML	The teacher wanted Print and a real export for the Score.	They live on the Lead Sheet toolbar, not in File, and not in the Score. She walked past them.

Master	Opened by all six; usable by none. The teacher would “close it rather than break the mix”; the teenager wanted “one 'make it sound finished' button.”	An empty rack, nine effects with no descriptions, “-inf LUFS” with no target, no preset chain.
Score	Excellent for the teacher. A wall for the teenager and a block for the musician.	It is the third tab for everyone. See §7 — a design question, not a bug.

7. Where it is too hard, too easy, or mis-sequenced

Too easy

- **The play-along passes an empty room.** Every other calibration problem is secondary to this one. At 17% per step, the loop rewards attendance, not accuracy.
- **The musician was made to prove what he plays nightly**, with no way to test out. *“Is there a way to skip ahead?”* There is — the step dots — but he found it only in the DOM.

Too hard

- **Ascend + Descend needs 16 strikes**, because it wants the top note played twice at the turnaround. Fifteen correct notes leave the exercise frozen with no indication why.
- **The Studio on arrival**, for the teacher and both teenagers. No orientation, no first step, no indication which track is theirs.
- **⌘-based instructions for a Chromebook classroom.** The only on-screen guidance in the Studio relies on a key the teacher's students do not have.
- **Score, for anyone who doesn't read notation**, and **Master, for everyone.**

Mis-sequenced

- **D $\flat$ (five flats) offered as lesson two** after C. *“No beginning-piano sequence goes C $\rightarrow$ D $\flat$.”* The *“pick next key center”* list runs chromatically, while the mode card lists keys by circle of fifths — two orders for the same choice.
- **Theory is ordered by brightness** (Lydian first). The teenager clicked Lydian first *“because it was the brightest card”* — the ordering is musically principled but not a learning path.

A genuine disagreement to decide, not fix

The teacher wants **Spelled** chord symbols (“C, Am”) and engraving rigour; she'd *“refuse to photocopy”* an empty measure. The musician wants **degree numbers** and found the Score an obstacle. The teenager wants neither and closed the Score in two seconds. These are three correct answers for three different people. The underlying question is whether the Score should be a peer tab offered to everyone, or something a user switches on because they read music.

8. What needs building or fixing, in priority order

Separated into **defects** (it exists and is wrong), **findability** (it exists and nobody could see it) and **missing** (it isn't there). They have very different costs.

#	CHANGE	KIND	WHAT IT UNLOCKS
1	Grade the play-along — notes hit, notes missed, and never "Nice work" for an empty performance.	Defect	Everything else. A teacher can assign; a student learns whether they got it. The data is already collected — only the completion check ignores it. What "passing" means is a product decision.
2	Live feedback while playing — light the next target, respond to a wrong note.	Missing	"Gets unstuck without me." The single most-cited frustration after grading.
3	A payoff line and a way back at the end of every activity: <i>"That's D Dorian — minor with a natural 6. It's the sound of the 2 chord in Midnight Groove."</i> + <i>"Back to your track."</i>	Missing	Closes the loop that all six personas said was open.
4	Give the Insight→Learn link a verb and a real size: <i>"Learn the scale behind this chord →"</i> .	Findability	The only bridge from a project into a lesson becomes visible.
5	Compute Insight's degree sentence from the key , not from the chord quality.	Defect	Removes a sentence that contradicts its own label on every non-tonic major chord.
6	Land the Practice Track on a one-purpose screen — chart, big Play, loop, tempo, <i>"play the C scale over this"</i> , Back to lesson — with the full Studio behind a link.	Missing	The hand-off delivers what its prompt promises. Rewrite the prompt in player language while there.
7	Fix PRISM Create's target track. Chords belong on the Chords track; the melody must survive; offer undo.	Defect	The teenager's favourite tool stops destroying work. Needs tracing — reported by one persona.
8	Say "Ionian = the major scale" in the lesson header, and give every mode card a one-line description in World Harmony's voice.	Missing	Removes the first jargon wall for two of three personas.
9	Fill in 11th and 13th chord symbols in <code>JAZZ_MAP</code> .	Defect	No more <i>"Gdom13"</i> . Small, and visible on every chart that uses them.

10	Fix the phantom bar and the rest-less staves in the notation engine.	Defect	DONE 20 SEP Six regression tests; verified on all three demos.
11	Give the Score Print/PDF and a MusicXML that contains the parts.	Missing	The teacher's route from "make something" to a printed part.
12	Give projects and lesson steps an address — a URL that reopens a project and one that opens step 4.	Missing	Assigning, refreshing, and not losing a rename.
13	Label the sidebar icons, PRISM, GROOVES and the # ruler toggle, and make a groove's name clickable.	Findability	Four of the most-loved features become discoverable. Cheap.
14	Re-sequence "next key" by circle of fifths, and fix Ascend + Descend's double top note.	Defect	Removes two avoidable walls for beginners.

Smaller items worth a ticket

- "Ionian Intervals: 1, 2, 3, 4, 5, 6, 7" should read "degrees" (or give real intervals).
- Plain triads print as "C maj" rather than "C"; Hybrid prints "D min9" with a space.
- The lesson's Notation view draws an undersized rest mid-bar in an empty bass staff instead of a centred whole-bar rest.
- Correct notes are drawn in **red** — which reads as "wrong" on a projector.
- The "Ready to start?" preview keyboard under "NOTE SEQUENCE" is blank.
- Lead Sheet slash view has no clef, key or time signature; turning Melody on shows the piano's grand staff.
- Lead Sheet's "+ Section" and "+ Repeat" were greyed out for a four-bar loop.
- Count-in bars are numbered "-1.1 -1.2 -1.3 -1.4".
- Mode cards are not buttons or links — not keyboard-reachable, not in the accessibility tree.
- "You completed the ionian lesson" — lower-case mode name mid-sentence.
- The lesson's back chevron is `history.back()`, so a lesson opened in a new tab backs out to a blank page.
- Insight's chip styling uses no `title`; hovering tells the user nothing.

9. Would they pay \$20 a month?

After their walkthroughs, each persona was asked the same question in character: what would make Music Atlas a no-brainer at \$20/month, which products they already use or would compare it to, what those do better today, and what they cost. Each was given its counterpart's findings so it could answer for the whole product, and told to verify competitor features and prices on the web rather than guess.

All three said no — today. The teacher: *“I can't justify it yet.”* The teenager: *“Today I'd pay nothing.”* The musician: *“For \$20 a month I get Moises Pro plus Chordify Premium, with change left over.”* Each named a short, specific list that would change the answer — and **the one item on all three lists is grading what the student actually played.**

About the prices below. They were gathered by the agents from the sources listed at the end of each section, on 20 September 2026. They were not independently re-checked for this report, and prices change. Where two agents disagreed — Chordify, reported as ~\$3.49/month by one and \$4 Basic / \$8 Premium by the other — both figures are shown.

How the three answers compare

	TEACHER	TEENAGER	MUSICIAN
Pay \$20/mo today?	No — <i>“can't justify it yet”</i>	No — \$0	No — would pay \$5–8
Would pay if...	It grades, assigns and prints	\$5 with one-click master, a share link and wrong-note feedback; \$20 only if it beats Simply Piano <i>and</i> BandLab	It analyses audio he brings in, fixes the lead sheet, and grades honestly
The missing core	The assign → submit → grade loop	Making something worth sharing	Analysing real recordings
Who's ahead there	Flat for Education, MusicFirst	BandLab, GarageBand, Simply Piano	Moises, Chordify, iReal Pro
What Music Atlas already wins on	Insight chord cards; the mode atlas; correct lesson notation	World Harmony; PRISM; Insight on <i>his own</i> song	The Insight card: degree + intervals + notes + plain English, linked to a playable lesson

9.1 The teacher

7th–9th grade general music and beginning piano. Budget framing: \$20/month is \$240 a year per teacher, “*more than I'd pay for any single tool below, so the bar is high.*”

What would make it a no-brainer, in her order:

1. **Scoring that means something** — notes hit and missed, timing accuracy, attempt history. *Unlocks a graded assignment instead of a participation checkbox. “This is the #1 blocker.”*
2. **Assign, roster, report** — assign “C Ionian Melody 1–6, 80% accuracy” to a section, see a class heatmap, export CSV. *Unlocks grading 14 students in minutes, and evidence for parents and administration. (/teacher exists but could not be tested in this environment.)*
3. **Print / PDF with parts.** *Unlocks worksheets, substitute plans, and a paper fallback when a Chromebook dies.*
4. **A real Practice Track screen** — chart, big Play, loop, tempo, “play C major over this”, Back to lesson. *Unlocks a projectable whole-class warm-up.*
5. **Fix the vocabulary** — “intervals” that are degrees, the “1st degree” captions, “C maj” for “C”, the double top note, ledger-line voicings, the phantom bar. *“I won't adopt a tool whose mistakes I'll have to un-teach.”*
6. **Circle-of-fifths key order**, and a key list she can set. *Unlocks differentiation: G and F for most, D ♭ for her two strong pianists.*

PRODUCT	WHAT IT DOES BETTER TODAY	REPORTED COST
Flat for Education	The full assign → submit → grade loop: composition and performance assignments, auto-graded theory and ear-training worksheets, Google Classroom / Teams / Canvas integration.	~\$99/teacher/yr + ~\$6/student. For her 14 students, ~\$183/yr — less than \$240, and it grades.
MusicFirst Classroom (with Auralia + Musition)	Deep graded theory and aural drills with a gradebook.	\$8/user/yr; the Auralia + Musition bundle ~\$4/user/yr (full cloud versions \$39 each, \$65 together)
Sight Reading Factory	Unlimited sight-reading generated from parameters. Music Atlas has one fixed exercise per key.	\$45/educator/yr; students ~\$3 each at 100+
Soundtrap for Education	Browser recording and creation, sold as a school licence by seat.	~\$4.98/seat/yr, 50-seat minimum (\$249)
MuseScore Studio	Full notation that prints and exports parts — “ <i>the paper artifact I need most.</i> ”	Free
Noteflight Learn	Browser notation, 500+ lessons and scores, a large premium score library.	\$69/yr for 10 users + \$2 per extra user

Where Music Atlas already wins: the Insight chord cards (*“once the captions are fixed, this is better teaching material than anything above”*); the mode atlas — diatonic, harmonic minor, melodic minor, harmonic major and double-harmonic families in all twelve keys, three clicks deep; correct lesson notation; real Score editing in the same app as the lessons; count-in, metronome and tempo control on the play-along.

What she'd tell her department head: *“I can't justify it yet. The mode and harmony material is better than what Flat or MusicFirst give us, but until a play-along stops saying 'Nice work' to a kid who played nothing, I can assign it to a class and pull a grade, and I can print a part, we'd be paying \$240 a year for something I can't grade, assign or hand to a sub — and Flat does all three for less.”*

Her sources: flat.io/edu/pricing · help.flat.io/en/education/plan · blog.flat.io (Flat vs Sight Reading Factory) · musicfirst.com/pricing · risingsoftware.com/shop/education · sightreadingfactory.com/pricing and its bulk-pricing article · musicednet.com/product/soundtrap · capterra.com (Noteflight pricing) · noteflight.com/learn · musescore.org · breezinthru.com (school plans). Quaver was not checked; no claim is made about it.

9.2 The teenager

15, self-taught, makes beats on his phone. Budget framing: “\$20 is a lot to a 15-year-old — most of a month of Spotify plus something.”

What would make it a no-brainer, in his order:

- 1. **A "make it sound finished" button.** “I want one button labelled 'Make it loud' and my thing sounds like a real song.” The difference between a project he hides and one he sends.
- 2. **A share link, not an export.** File offers Export Audio, Export MIDI, Export Lead Sheet — and “Export Analysis JSON”, “sitting in the same little menu as the stuff I need.” No share link, no vertical-video clip. “A .wav in my Downloads folder is not 'showing a friend' in 2026.”
- 3. **Real songs he actually likes** — search a song, get the chords, play along slowed down. “That's the whole reason my friends open an app.”
- 4. **GROOVES and PRISM on the front door**, not in unlabelled bottom tabs — “I'd mess around for an hour.”
- 5. **VIBES and STYLES at the top of Insight**, not under “Built on the 1st degree of the major scale” and “C Melodic Minor 89%.”
- 6. **Stop losing his work.** “If it loses a name, I don't trust it with a song.”

PRODUCT	WHAT IT DOES BETTER TODAY	REPORTED COST
GarageBand (iOS)	Loops, drummer, amps, a one-tap finished sound, and export straight to camera roll or Messages. “Already on my phone.”	Free
BandLab	The Studio's idea in browser and phone: 16 tracks free, 160k+ royalty-free sounds, built-in auto-mastering, every project a link, a social feed.	Free core; Membership ~\$14.95/mo or \$99–199/yr
FL Studio Mobile	What his beat-making friends actually use. A real sequencer, real sounds, no subscription.	\$15, once
Moises	Upload any song: stems split, chords detected, tempo slowed. “‘Get better at a song I like', which Music Atlas doesn't do at all.”	Free (5/mo); \$3.99/mo Premium
Chordify	Chords for almost any song, instantly.	~\$3.49/mo (the musician reported \$4–8; see note)
Ultimate Guitar	Tabs and chords for everything, with playback.	\$9.99/mo, \$49.99/yr
Simply Piano	Listens and says right or wrong in real time, with songs he recognises. “Music Atlas's Hold exercise gave me zero feedback.”	\$17.90/mo, \$169.90/yr

Yousician / flowkey	Real-time note detection, a real song library, structured levels.	Each \$19.99/mo (~\$9.99/mo billed annually)
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The comparison that decides it: *“At \$20/month Music Atlas is asking Simply Piano / Yousician / flowkey money — but it doesn't listen to me the way they do, and it doesn't have their song libraries. And as a beat-maker it's asking money for something BandLab and GarageBand give me free and FL Studio Mobile sells once for \$15.”*

Where Music Atlas already wins: World Harmony (*“something none of those apps have”*); PRISM (*“a better idea than anything in Simply Piano — it makes music instead of grading me”*); Insight telling him the key and scale of *his own* song, where Chordify and Moises only analyse other people's; and being one app that both teaches and produces — *“nothing on my phone does both.”*

Would he pay? *“No. Today I'd pay nothing, because in 60 seconds in the Studio I couldn't make one thing I'd want anyone to hear, and every word on screen was for someone older than me.”* At **\$5/month** he'd consider it with a one-click finish, a share link and wrong-note feedback. At \$20 it would have to beat Simply Piano at teaching *and* BandLab at making. *“It's genuinely close to being that; it's just that the part a teenager wants (GROOVES, PRISM, VIBES) is hidden behind the part a teacher wants.”*

His sources: pianostartguide.com (Simply Piano review) · account.yousician.com/plans · help.flowkey.com (subscription plans) · bandlab.com/membership-pricing and [/creation-features](https://bandlab.com/creation-features) · stemsplit.io (Moises review) · chordify.net/premium · help.ultimate-guitar.com (pricing) · App Store listings for FL Studio Mobile and GarageBand.

9.3 The gigging musician

34, fifteen years of bass and keys on function and soul gigs. Wants the names for what his hands already do.

What would make it a no-brainer, in his order:

1. **Analyse real recordings** — drop in the MP3 a bandleader sent and get the key and changes as “2 min9 · 5 dom13”, with interval chips. *Learning a 40-song wedding set in an evening instead of a week.* He saw only MIDI demos and did not test audio import, so this is “I didn't find it”, not “it's missing”.
2. **A gig-ready lead sheet** — “G13, not Gdom13. Dm9, not D min9.” Working repeats and sections, big symbols, eight bars a line, a clean PDF. *Charting a tune for a dep in ten minutes; the Numbers view then works in any key.*
3. **Grade what he actually plays.** “For me that means the lesson data is worthless.” *Woodshedding a weak key and trusting the score.*
4. **Close the loop** from Insight to a lesson built on *this* chord in *this* key, and back. *Finally having the word when a bandleader calls “Dorian vamp on the 2”.*
5. **Call-the-changes ear drills** — hear a progression, answer “1–6–2–5”. “*This is literally the gig skill.*”
6. **Transpose-and-play-along on any chart** — backing band, twelve keys, speed trainer.

PRODUCT	WHAT IT DOES BETTER TODAY	REPORTED COST
iReal Pro	A huge chart library; a backing band in any key; spelling that stays correct when transposed; and it already has "Number Notation" . “ <i>It wins today for charts and play-along, by miles.</i> ”	\$21.99 once (iOS); style packs \$6.99
Moises	Stem separation, chord and key detection, transpose to all twelve keys, click track. “ <i>What I'd actually use to learn a set.</i> ”	\$3.99/mo Premium, \$9.99/mo Pro
Chordify	Chords from YouTube or an upload, synced to playback, with transpose and capo. “ <i>Its chord names are rougher than yours.</i> ”	\$4/mo Basic, \$8/mo Premium
Hookpad / TheoryTab	Already labels chords degree-first with the letter second , colour-coded, over tens of thousands of analysed songs. “ <i>So I have to take back 'the first time software wrote chords the way I say them.'</i> ”	\$7.99/mo or \$199 lifetime
Band-in-a-Box	Realistic backing tracks for any changes typed in.	From \$129 (Audiophile \$669)
Transcribe! · Amazing Slow Downer	Slow down and loop real audio — essential for transcribing. “ <i>Music Atlas has nothing equivalent that I saw.</i> ”	~\$39 · ~\$50 (older secondary sources)
Ultimate Guitar Pro	Official tabs and pop/rock charts.	\$9.99/mo or \$59.99/yr (sources conflict)

Where Music Atlas already wins: *“It's not the numbers by themselves. It's the Insight chord card.”* The hybrid label with the letter name greyed beside it, interval chips (“R b3 5 b7 9”), the note spelling, a plain-English line — all worked out *from the track he's in*, and linked to a piano-roll lesson he can play. *“None of the products I checked turns 'what's in my track' into 'what the extension is called, by interval' and then 'go play it' in one place. iReal and Hookpad label chords. They don't teach me what the 9 or the b7 is doing. That card, plus a lesson that grades honestly, is the product.”*

Would he pay? No. *“Today I'd pay \$5–8/month, about Hookpad money, for the Insight card and the degree view, knowing it's a learning toy.”* The shortest list that flips it to yes: (1) analyse audio he brings in, through the same Insight card; (2) fix the lead sheet; (3) grade honestly and link the lesson back to the chord. *“With those three, it's the only tool that does Moises's job and teaches me the word, and \$20 is fair. Without #1, it stays a \$5 theory toy.”*

His sources: App Store and irealpro.com (iReal Pro, Number Notation) · guitarchalk.com (Chordify review, UG Pro review) · stemsplit.io and moises.ai/features (Moises) · hooktheory.com (Hookpad pricing and guide) · pgmusic.com (Band-in-a-Box) · seventhstring.com (transcription tools). He could not verify a product called “Nail the Number System” and makes no claim about it.

9.4 What the three answers add up to

1. One fix is on every list. Grading what was actually played is the teacher's #1 blocker, the teenager's condition for paying even \$5, and the musician's third of three conditions. It is also, from Part 1, the one whose data already exists in the code. It is the highest-leverage change in this study by a wide margin.

2. The three personas want three different products. The teacher wants a gradebook with an LMS integration. The teenager wants a social beat-maker. The musician wants an audio analyser and chart tool. Each already has a specialist that does its job — Flat, BandLab, Moises — and each of those is cheaper than \$20 a month, or free, or one-time.

3. The shared core is the Insight card. Asked independently where Music Atlas wins, all three pointed at the same thing: naming what is inside the music — degree, intervals, notes, a plain sentence — and handing the user something to play. None of their competitors does that end to end. It is the product's genuine differentiator, and today it is partly wrong (Part 1, §4) and poorly connected to the lessons (§5.3–5.4).

4. \$20 a month puts Music Atlas against its weakest comparison. The consumer apps at that price — Simply Piano, Yousician, flowkey — are strongest precisely where Music Atlas is weakest today: listening to the player in real time and offering a library of songs people know. Most of what the personas actually use costs \$4–10 a month or is paid for once.

5. A single monthly price may not fit the teacher at all. Every education product she checked is sold per seat, per year, to a school, often through a purchase order — not per teacher per month. Her comparison was never "\$20 against \$15"; it was "\$240 against \$183, and theirs grades."

Read these conclusions for what they are. They come from three simulated personas, each built from a short brief, reasoning about pricing they researched themselves. They are a structured way to surface hypotheses — which features gate willingness to pay, and against which competitors — not evidence of what real teachers, teenagers or musicians would pay. Testing them with real people would be the next step.

Appendix A — Words the teenagers did not know

Both teenager personas kept a running list of every word, symbol or button they didn't understand, including ones they guessed right. It is reproduced here because it is a direct inventory of the vocabulary wall a self-taught beginner hits.

Guessed, roughly right

BPM · M / S · ► · Quantize · "min / maj" · 4/4

No idea, or ignored

Ionian · Dorian · Lydian · Mixolydian · Phrygian · mode · Diatonic / Relative / Parallel Modes · "Ionian Intervals: 1, 2, 3, 4, 5, 6, 7" · "Dorian Intervals: 1, 2, ♭3, 4, 5, 6, ♭7" · key center · curriculum · Hold vs Play Along · Demo vs Practice vs Play Now · "sequence" · Prism · 2–5–1–6 · 1–5–6–4 · 1 maj · 6 min · 5 dom7 · 1 maj7 · 1 maj/3 · min9 · dom13 · maj9 · min7 · R 3 5 · R b3 5 b7 9 13 · "Built on the 1st degree of the major scale" · "Also found in 12 scales" · "Parent: G Ionian" · "Key of C 85%" · "C Melodic Minor 89%" · Triads · 7th Chords · Inversions · ♭ · KEY: C · 1:1:1 · MON · RELEASE · –1.9 · AUTO · MASTER BUS · FX · Record Arm · Automation · Snap to Grid · Chord Ruler: Note Names · Count In · Lock root note · Grid 1/16 · VEL · VELOCITY · "82 notes 201%" · RDE CRS TM1 TM2 FLT OHH PHH · SAMPLE · Gain Match · Bypass · LUFS · –inf · IN/OUT · RETURN A/B · OTT · De-Esser · Saturator · Presence · Compressor · Gate · EQ · 0 dB · PALETTES · Barlines · D.C. · D.S. · D.C./D.S. al Coda · ARTICULATIONS · ACCIDENTALS · RHYTHMIC · TIE · SLUR · C7 CHORD · Hybrid · Replace / Locked / Merge · + Section · + Repeat · MusicXML · "chord lane" · ⌘-drag · Analyze Chords · Contour: Arch · Range A4—E5 · Feel: straight · Strum Mode: Synchronized · Swing · VIBES / STYLES · "friend's Compass" · the © and ~ track icons · every icon in the left sidebar

Reading the list. Much of it is professional DAW vocabulary a beginner neither needs nor expects to know, and that is fine in the Studio. What matters is the Learn half: *Ionian*, *Hold vs Play Along*, *Demo vs Practice vs Play Now* and *curriculum* all appear in the path a beginner is meant to walk, and none is explained where it appears.

Appendix B — Where the evidence lives

All scripts and screenshots are in the research session's scratch directory, named by persona and step: `teacher-a-*`, `teacher-b-*`, `teen-a-*`, `teen-b-*`, `player-a-*`, `player-b-*` (A = Learn → Studio, B = Studio → Learn). No repository files were modified by the research agents.

RUN	SCRIPTS	SCREENSHOTS
Teacher · Learn → Studio	23	38
Teacher · Studio → Learn	21	42
Teenager · Learn → Studio	23	32
Teenager · Studio → Learn	22	30
Musician · Learn → Studio	20	37

Musician · Studio → Learn	15	30
Total	124	209

Code references are to the working tree of branch `Peter` on 20 September 2026 and may shift as that branch changes.